



Foundations of Finance



Financial Statements and Free Cash Flow



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Partial Review of the Last Class

- ▶ A project costs \$20M today and \$20M in a year, and returns \$30M at the end of years 2 and 3. At $r = 8\%$, should the firm take it?
- ▶ Why does $\text{NPV}(8\%, C_0 : C_3)$ give 10.20 instead of 11.02?
- ▶ The project's IRR is 22.47%. What is the IRR of a bond called?
- ▶ Name two cases in which the IRR rule can mislead
- ▶ What does the payback rule ignore?

Introduction

“To our shareholders:

Our ultimate financial measure, and the one we most want to drive over the long-term, is free cash flow per share.

Why not focus first and foremost, as many do, on earnings, earnings per share, or earnings growth? The simple answer is that earnings don’t directly translate into cash flows, and shares are worth only the present value of their future cash flows, not the present value of their future earnings. Future earnings are a component—but not the only important component—of future cash flow per share. Working capital and capital expenditure are also important, as is future share dilution.”

—Jeff Bezos, 2004 letter to shareholders

Today

- ▶ Cash flow definition and core principles
- ▶ Introduction to accounting statements
- ▶ Accounting adjustment #1: depreciation
- ▶ Accounting adjustment #2: working capital
- ▶ Example: free cash flow at Apple Inc.

Cash Flow Fundamentals

- ▶ The value of an asset is the present value of its cash flows

$$\text{Cash flow} = \$ \text{received} - \$ \text{paid}$$

- ▶ Essential rules for computing cash flows:
 - ▶ Use cash flows, not accounting earnings
 - ▶ Earnings do not reflect the true timing of cash flows
 - ▶ Use after-tax cash flows
 - ▶ Cash flows should only reflect operating activities
 - ▶ Equivalent to the cash flows of an all-equity financed firm
 - ▶ Ignore financing decisions (debt, dividend timing, etc.)
 - ▶ These will be reflected in the discount rate (cost of capital)

Computing Cash Flows

- ▶ We want to compute

$$CF = \text{Total Cash Inflows} - \text{Total Cash Outflows}$$

- ▶ Start by connecting inflows and outflows to firm accounting statements

$$CF = \left(\begin{array}{c} \text{Operating} \\ \text{Revenues} \end{array} \right) - \left(\begin{array}{c} \text{Operating} \\ \text{Costs} \end{array} \right) - \left(\begin{array}{c} \text{Capital} \\ \text{Expenditures} \end{array} \right) - \left(\begin{array}{c} \text{Income} \\ \text{Taxes} \end{array} \right)$$

- ▶ Operating revenues: main inflow/sales from primary operations
- ▶ Operating costs: cost of goods sold (“COGS”) + selling, general, and administrative expenses (“SG&A”)
- ▶ Capital expenditures (“CapEx”): investments in long-term capital assets
- ▶ Income taxes: paid on corporate profits (earnings)

Accounting Statements

- ▶ We need to compute cash flows from accounting statements
 - ▶ Standard form in which firm-level data are available
 - ▶ Accountants' goal: inform about a firm's overall profitability, not its value
- ▶ Principles governing accounting statements:
 - ▶ **Accrual accounting:** record revenues and costs when products and services are sold or provided, not when they are paid for
 - ▶ **Operating vs. capital expenditures:** only expenses that create revenue in the current period are counted as operating expenses. Investments that create revenues over several periods are written off over those periods
- ▶ To translate accounting earnings into cash flows, we need to:
 - ▶ Add back non-cash expenses (e.g., depreciation)
 - ▶ Subtract cash outflows that are not “expensed” (capital expenditures)
 - ▶ Convert accrual timing to actual timing using changes in working capital

A stylized income statement

Revenues (Sales)

- Cost of Goods Sold (COGS)
 - Sales, General, and Administrative (SG&A)
-

= Earnings Before Interest, Taxes, Depreciation, and Amortization (EBITDA)

- Depreciation & Amortization (D&A)
-

= Earnings Before Interest and Taxes (EBIT, a.k.a. Operating Profit)

- Interest Expense
-

= Earnings Before Taxes (EBT)

- Taxes (at effective rate t)
-

= Net Income (a.k.a. Earnings)

A Stylized Balance Sheet

Assets		Liabilities	
(D)	Marketable Securities	Accounts Payable	(O)
(O/D)	+ Cash and Cash Reserves	+ Short-Term Debt	(D)
(O)	+ Accounts Receivable		
	= Current Assets	= Current Liabilities	
(O)	Gross Property, Plant, and Equipment (PPE)	+ Long-Term Debt	(D)
(O)	– Accumulated Depreciation		
	= Net Property, Plant, and Equipment	= Total Liabilities	
		+ Stockholders' Equity	(E)
	Total Assets	Total Liabilities and Stockholders' Equity	

(O) Operating Assets and Liabilities – (D) Debt Assets and Liabilities – (E) Equity

Adjustment 1: Depreciation

- ▶ Accounting earnings do not deduct capital expenditures when they occur
 - ▶ Designed to capture profitability of a firm, used to compute corporate taxes
 - ▶ Large investments often imply low cash flows, but do not imply losses
- ▶ Spread capital expenditure over multiple periods (“depreciation”)
 - ▶ Idea: profits net out the cost of replacing deteriorating capital
 - ▶ “Straight-line” depreciation: equal depreciation per year over fixed period
- ▶ Why should we care about depreciation if it is not a real outflow?
 - ▶ If we have earnings data, need to “undo” depreciation to get to cash flows
 - ▶ Taxes are based on accounting income, so the same investment will generate different after-tax cash flows depending on how it is depreciated

Corporate Taxes

- ▶ Taxes are paid on operating profit, defined by accountants as

$$\text{Operating Profit} = \text{Operating Revenues} - \text{Operating Costs} - \text{Depreciation}$$

- ▶ If we define t to be the effective tax rate, then taxes are approximated by

$$\text{Income Taxes} = t \times \text{Operating Profit}$$

- ▶ Substitute into our cash flow formula to obtain

$$CF = (1 - t) \times \text{Operating Profit} - \text{Capital Expenditures} + \text{Depreciation}$$

- ▶ Or, using EBITDA (= Operating Revenues – Operating Costs)

$$CF = (1 - t) \times EBITDA - \text{Capital Expenditures} + t \times \text{Depreciation}$$

Ex: Depreciation

- ▶ You are considering purchasing an assembly line for \$100M
 - ▶ Generates \$40M per year for 10 years and costs \$25M per year to operate
 - ▶ Depreciated over 10 years using straight-line depreciation
 - ▶ Corporate tax rate is 20%

Ex: Depreciation

- Project earnings and cash flows (using operating profit):

The diagram illustrates the calculation of cash flows from operating profit. Red arrows point from labels to specific parts of the table: 'Revenues' points to the \$40M in Year 1; 'Tax Rate' points to the 0.2 in the Year 1 calculation; 'Depreciation' points to the \$10M in Year 1; and 'Costs' points to the \$25M in Year 1.

Year	Operating profit (before taxes)	Operating profit (after taxes)	Cash flow
0	\$0	\$0	-\$100M
1	$\$40\text{M} - \$25\text{M} - \$10\text{M} = \5M	$(1 - 0.2) \times \$5\text{M} = \4M	$\$4\text{M} + \$10\text{M} = \$14\text{M}$
2	\$5M	\$4M	\$14M
3	\$5M	\$4M	\$14M
...	...	...	...
10	\$5M	\$4M	\$14M

Ex: Depreciation

► Project earnings and cash flows (using EBITDA):

Revenues

Tax Rate

Depreciation

Year	EBITDA	EBITDA after taxes (a.k.a. “EBIDA”)	Cash flow
0	\$0	\$0	−\$100M
1	$\$40\text{M} - \$25\text{M} = \$15\text{M}$	$(1 - 0.2) \times \$15\text{M} = \12M	$\$12\text{M} + 0.2 \times \$10\text{M} = \$14\text{M}$
2	\$15M	\$12M	\$14M
3	\$15M	\$12M	\$14M
...	...	...	...
10	\$15M	\$12M	\$14M

Costs

Handling Interest Payments

- ▶ Accountants consider interest payments as an operating expense
 - ▶ Deduct them from net income (earnings)
- ▶ But interest is a financing expense, not related to operations
 - ▶ Simply payments from one group of investors to another
 - ▶ We ignore financing activities when computing free cash flows, which measure total inflows and outflows at the firm level
- ▶ As with depreciation, this accounting choice matters for corporate taxes
 - ▶ Because taxes are paid on operating profits net of interest, each \$1 of interest reduces taxes by $\$t$
 - ▶ This “tax shield” is a real source of income that must be accounted for
 - ▶ We will ignore it for now and handle it later with an adjustment to the discount rate (the “WACC”)

Adjustment 2: Working Capital

- ▶ Accountants measure revenues and costs when a sale is recorded (“accrual” accounting)
 - ▶ This may not be when the cash is actually received!
 - ▶ These timing differences are measured by changes in working capital

$$\left(\begin{array}{c} \text{Net Working Capital} \\ \text{NWC} \end{array} \right) = (\text{Inventory}) + \left(\begin{array}{c} \text{Accounts} \\ \text{Receivable} \end{array} \right) - \left(\begin{array}{c} \text{Accounts} \\ \text{Payable} \end{array} \right)$$

- ▶ **Inventory:** “Cost of goods sold” ignores cost of items produced but not yet sold
 - ▶ When inventory is rising, cost of good sold understates cash outflows
 - ▶ When inventory is falling, cost of goods sold overstates cash outflows
- ▶ **Accounts receivable:** Money owed to a firm by its customers
 - ▶ Accounting sales overstate cash inflows if the firm has not yet been paid
 - ▶ Understate cash inflows if the firm is receiving payment for past sales
- ▶ **Accounts payable:** Opposite of accounts receivable (money owed to suppliers)

Ex: Changes in NWC

- ▶ When net working capital changes, we need to **subtract it** to get cash flows
- ▶ Imagine you run a retail shoe chain.
 - ▶ This quarter, you buy 2M pairs of shoes for \$30 each
 - ▶ For the each of the next two quarters, you sell 1M pairs for \$80 each
 - ▶ Corporate tax rate is 20%

Cash flows (in \$ million)

Qtr	After-tax profits (accrual accounting)	After-tax cash flow (true timing)	Inventory
0	\$0	−\$60	\$60
1	$(\$80 - \$30) \times (1 - 0.2) = \$40$	$\$80 - 0.2 \times (\$80 - \$30) = \70	\$30
2	\$40	\$70	\$0

- ▶ Subtracting the change in inventory from accounting profits → cash flow

Cash Flow Formulas with NWC

- ▶ A change in NWC affects the company's cash flow
 - ▶ Increase in working capital means decrease in cash flow
 - ▶ Decrease in working capital means increase in cash flow
- ▶ We can adjust our cash flow formula to account for this (“Δ” means “change in”)

$$CF = (1 - t) \times \text{Operating Profit} - \text{Capital Expenditures} + \text{Depreciation} - \Delta NWC$$

- ▶ Alternatively, use the EBITDA version

$$EBITDA = \text{Operating Revenues} - \text{Operating Costs}$$

$$CF = (1 - t) \times EBITDA - \text{Capital Expenditures} + t \times \text{Depreciation} - \Delta NWC$$

How to Handle Cash

- ▶ Should cash be included in the firm's operating NWC?
- ▶ Sometimes, if:
 - ▶ It is required for the day-to-day operations of the firm, and varies spontaneously with the firm's operating activities
 - ▶ In this case, an increase in cash “locks up” resources, creating a negative cash flow
 - ▶ Conversely, a decrease in required cash “frees” resources for investors
- ▶ However, if a firm has excess cash or cash reserves (liquidity), this should not be included in the FCF calculation
 - ▶ We will separately add this (and other financial assets) when valuing the firm
- ▶ In practice, operating NWC is generally defined as the difference between non-cash current assets and non-interest-bearing current liabilities

Example: Apple's Cash Flow

► Let's compute Apple's cash flow in 2024.

Item (\$ millions)	2024
Revenue (Sales)	391,035
– COGS excluding D&A	198,907
– SG&A Expense	57,467
= EBITDA	134,661
– Depreciation & Amortization Expense	11,445
= Operating Profit (EBIT)	123,216
– Non Operating Income/Expense	269
– Interest Expense	-
= Pretax Income (EBT)	123,485
– Income Tax (@ effective rate of 24%)	29,749
= Net Income	93,736

Apple's Balance Sheet

Assets (\$ millions)	2024	2023		Liabilities (\$ millions)	2024	2023
Cash & ST Investments	65,171	61,555		ST Debt & Current LT Debt	22,511	17,382
+ Accounts Receivable	66,243	60,985		+ Accounts Payable	68,960	62,611
+ Inventories	7,286	6,331		+ Income Tax Payable	26,601	-
+ Other Current Assets	14,287	14,695		+ Other Current Liabilities	58,320	65,315
= Total Current Assets	152,987	143,566		= Total Current Liabilities	176,392	145,308
Gross PP&E	129,362	125,260		+ Long-Term Debt	96,548	106,548
– Accumulated Depreciation	73,448	70,884		+ Risks & Charges	9,254	15,457
= Net PP&E	55,914	54,376		+ Deferred Taxes	-19,499	19,454
				+ Other Liabilities	25,836	3,670
Other LT Investments	91,479	100,544		= Total Liabilities	308,030	290,437
Other Assets	45,101	54,097		Total Equity	56,950	62,146
Total Assets	364,980	352,583		Total Liabilities & Equity	364,980	352,583

Apple's Cash Flow

- ▶ Cash flow computation (in \$ billions):

$$\begin{aligned} CF &= (1 - t) \times \text{Operating Profit} + \text{Dep\&Amort} - \text{CapEx} - \Delta NWC \\ &= (1 - 0.24) \times 123.2 + 11.4 - \text{CapEx} - \Delta NWC \end{aligned}$$

- ▶ Capital expenditures:

$$\begin{aligned} \text{CapEx} &= \Delta \text{Gross PPE} \\ &= (129.3 - 125.3) \\ &= 4.1 \end{aligned}$$

Apple's Cash Flow

► Working Capital

ΔNWC

$$\begin{aligned} &= \Delta \text{Accounts Receivable} + \Delta \text{Inventory} - \Delta \text{Accounts Payable} \\ &= (66.2 - 61.0) + (7.3 - 6.3) - (69.0 - 62.6) \\ &= -0.1 \end{aligned}$$

Apple's Cash Flow

- ▶ Plugging in for capital expenditures and operating net working capital:

$$\begin{aligned} CF &= (1 - t) \times \text{Operating Profit} + \text{Dep\&Amort} - \text{CapEx} - \Delta NWC \\ &= (1 - 0.24) \times 123.2 + 11.4 - 4.1 - (-0.1) = 101.1 \end{aligned}$$

- ▶ Equivalent formula using EBITDA:

$$\begin{aligned} CF &= (1 - t) \times EBITDA + t \times \text{Dep\&Amort} - \text{CapEx} - \Delta NWC \\ &= (1 - 0.24) \times 134.7 + 0.24 \times 11.4 - 4.1 - (-0.1) = 101.1 \end{aligned}$$

- ▶ Use whichever version of the formula you find more intuitive

Apple's Cash Flow

- ▶ For comparison, Apple's accounting earnings (net income) is only \$93.7B
 - ▶ Why is free cash flow so much higher than net income?

- ▶ Can rearrange our earlier definitions to obtain

$$\begin{aligned} CF &= NI + (1 - t) \times \text{Interest Payments} + \text{Dep\&Amort} - \text{CapEx} - \Delta NWC \\ &= 93.7 + (1 - 0.24) \times 0 + 11.4 - 4.1 - (-0.1) = 101.1 \end{aligned}$$

- ▶ Apple's cash flow is higher mostly because CapEx is smaller than Depreciation & Amortization
 - ▶ Actual long-term investments are below accountants' estimated depletion
 - ▶ Lower investment requires less cash, leaving more resources for investors today
 - ▶ In principle could reflect lower growth rather than higher profit

Summary

- ▶ Free cash flow measures the total net resources flowing out of the firm to investors
 - ▶ Correct object to value the firm using DCF
- ▶ We can compute free cash flows from accounting statements
- ▶ But need to adjust measures for accounting timing
 - ▶ Depreciation: fictitious expense (that really affects taxes)
 - ▶ Capital expenditure: actual expense ignored by accountants
 - ▶ Operating working capital: corrects accrual timing to reflect when revenues and costs are actually paid